



Compiler Optimization Report - YOLOv8s Instance Segmentation

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1 Summary

For YOLOv8s Instance Segmentation at 1088x1920x3 resolution:

Performance simulation estimates a Total Processing Time (for the compiled ONNX file) of

- 8.40 ms, which corresponds to
 - 73.5% ACE utilization, using 24 Analog Compute Engines.

The latency estimate excludes Digital NPU processing time, which is not yet supported for this network.

There are no digital bottlenecks in the Analog NPU processing time.

Automatic optimization performs as well as the best manual optimization.

We may be able to improve performance in the future by finding a more optimal placement of weight blocks to ACEs (to increase the ACE utilization), but note that the current ACE utilization of 73.5% is already very high.

We may be able to improve performance in the future by allowing weight block widths to exceed 256. The hardware design recently increased support to 512, and the compiler will add support for 512 in the future.

The estimated power is

- 0.4 Watts at the target frame rate (30 fps), and
- 1.3 Watts at the maximum frame rate (119 fps).

Power estimates are approximate, and will be more precise in the future.

2 Reproducing the results

The results in this report were produced using the `“high_optimization”` compiler config for YOLOv8s Instance Segmentation.

Note that some variance in results is expected (up to a few percent), due to non-deterministic, multi-threaded compiler optimizations.

The `mythic-compiler` command was

```
mythic-compiler \  
  --input-artifact $MYTHIC_SDK_ROOT/models/training/yolov8/yolov8s-  
seg_1920x1088_trained.tar.gz \  
  --compiler-config $MYTHIC_SDK_ROOT/mythic-model-zoo/configs/yolov8/compiler/  
yolov8s_seg_1088x1920x3_m2024_high_optimization.yaml \  
  --output-artifact /rscratch/tonbomythic3/<your_username>/  
yolov8s_seg_1088x1920x3_m2024_high_optimization_compiled.tar.gz
```

The `mythic-ppa-estimators` command for the maximum frame rate (119 fps) was

```
mythic-ppa-estimators \  
  --estimate-performance \  
  --estimate-power \  
  --estimate-latency
```

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```
/rscratch/tonbomythic3/<your_username>/  
yolov8s_seg_1088x1920x3_m2024_high_optimization_compiled.tar.gz
```

The `mythic-ppa-estimators` command for the target frame rate (30 fps) was

```
mythic-ppa-estimators \  
  --estimate-performance \  
  --estimate-power \  
  --power-inference-rate 30 \  
  /rscratch/tonbomythic3/<your_username>/  
  yolov8s_seg_1088x1920x3_m2024_high_optimization_compiled.tar.gz
```

3 Performance estimates

The `mythic-ppa-estimators` tool outputs the following performance statistics:

```
Analog NPU Total Estimated Processing Time for Compiled ONNX File: 8.40 ms  
Analog NPU Estimated Frame Rate for Compiled ONNX File: 119.04 fps (frames per second)  
Critical Path ACE Latency: 8.40 ms  
Maximum SRAM Read/Write Time (over 6 ACE tiles): 7.20 ms  
Maximum SIMD Operation Time (over 6 ACE tiles): 0.00 ms  
Average SRAM Read/Write Time (averaged over 6 ACE tiles): 5.64 ms  
Average SIMD Operation Time (averaged over 6 ACE tiles): 0.00 ms  
Total Executed ACE Operations: 926,432  
Total Executed ACE MACs: 133,839,667,200  
Maximum Theoretical ACE Execution Time (With No Parallelization Using 1 ACE): 148.23 ms  
Minimum Theoretical ACE Execution Time (With Even Parallelization Across 24 ACEs): 6.18  
ms  
ACE Utilization (Minimum Theoretical Time/Estimated Processing Time): 73.52%  
Total Estimated MACs Targeting Digital NPU: [TBD]  
Total SRAM Bytes Read Across Chip: 2,434,269,432 bytes  
Total SRAM Bytes Written Across Chip: 1,207,817,640 bytes  
Estimated Die Area to Achieve Estimated Processing Time: 126.55 mm^2
```

NOTE: The information reported above may include the Analog NPU or the Analog NPU and a digital NPU as indicated. This estimated processing time and associated frame rate only covers the execution of the compiled ONNX model(s) running on one or both of those compute solutions in isolation. It does not include PCIe transfer overhead. Performance estimation for additional processing steps and a combined compute hardware solution will be added in future versions.

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4 Power estimates

The `mythic-ppa-estimators` tool outputs the following power estimates.

Power at the target frame rate (30 fps):

```
Process Nodes: Analog 28nm, Digital 5nm
Number of ACEs: 24
Inference Rate: 30 frames/second (fps)
```

```
Analog NPU Estimated Power: 0.387 W for target ONNX file
  Functional Unit Power: 0.312 W
  Interconnect Power: 0.075 W
```

NOTE: Estimation for leakage, clock tree, chip I/O power will be added in future versions.

Power at the maximum frame rate (119 fps):

```
Process Nodes: Analog 28nm, Digital 5nm
Number of ACEs: 24
Inference Rate: 119 frames/second (fps)
```

```
Analog NPU Estimated Power: 1.347 W for target ONNX file
  Functional Unit Power: 1.049 W
  Interconnect Power: 0.298 W
```

NOTE: Estimation for leakage, clock tree, chip I/O power will be added in future versions.

5 Compiler reports

5.1 L0 IR stats

From near the end of mythic-compiler stdout:

```
===== L0 IR stats =====  
      Tiles: 7  
      Buffers: 173  
      Launchers: 267  
      ACE Launchers: 113  
      ACE Weights: 18927664  
      ACE calculations: 926432  
      Max ACE calculations: 161840 on tile 2_2  
=====
```

5.2 ACE utilization report

The compiler-generated `ace_utilization.txt`:

Summary:

- 87.32% ACE utilization is predicted by the compiler,
- assuming no digital bottlenecks (which can typically be eliminated).
- 926,432 ACE calculations are spread across 24 ACEs, which is
- 38,601.33 ACE calculations per ACE (on average).
- 44,205 ACE calculations are in the critical path, implying
- 7.07 ms predicted inference time (160 ns per ACE calculation).

Tile	ACEs	ACE calculations	ACE calculations Per ACE	Predicted Full Network ACE Critical Path	Predicted ACE Utilization
1_1	0	N/A	N/A	N/A	N/A
3_1	4	153,408	38,352.00	44,205	86.76%
4_1	4	153,000	38,250.00	44,205	86.53%
1_2	4	149,736	37,434.00	44,205	84.68%
2_2	4	161,840	40,460.00	44,205	91.53%
3_2	4	154,224	38,556.00	44,205	87.22%
4_2	4	154,224	38,556.00	44,205	87.22%
All	24	926,432	38,601.33	44,205	87.32%

5.3 Weight utilization report

The compiler-generated `weight_utilization.txt`:

Summary:

- 63.34% of analog weight storage capacity is used, using
- 18,927,664 weights out of 29,884,416 capacity, using
- 113 weight blocks.

Tile	Weight Blocks	Allocated Weights	Weight Capacity	Utilization
3_1	19	2,996,384	4,980,736	60.16%
4_1	18	3,380,928	4,980,736	67.88%
1_2	20	2,880,336	4,980,736	57.83%
2_2	16	2,748,480	4,980,736	55.18%
3_2	22	3,665,904	4,980,736	73.60%
4_2	18	3,255,632	4,980,736	65.36%
All	113	18,927,664	29,884,416	63.34%

5.4 SRAM utilization report

The compiler-generated `sram_utilization.txt`:

Summary:

- 5,157,699 bytes of SRAM is used, consisting of
- 4,818,688 bytes **for** buffers (93.43%) and
- 339,011 bytes **for** control flow (6.57%), using
- 20.49% of the simulated on-chip SRAM capacity.

Tile	Buffer Allocation (bytes)	Control Flow Allocation (bytes)	Total Allocation (bytes)	SRAM Capacity (bytes)	Utilization
3_1	1,189,216	78,111	1,267,327	4,194,304	30.22%
4_1	214,144	31,844	245,988	4,194,304	5.86%
1_2	422,112	69,743	491,855	4,194,304	11.73%
2_2	592,224	42,556	634,780	4,194,304	15.13%
3_2	544,032	82,830	626,862	4,194,304	14.95%
4_2	1,856,960	33,927	1,890,887	4,194,304	45.08%
All	4,818,688	339,011	5,157,699	25,165,824	20.49%